

MARION ISLAND, South Africa

WMO: 689940

Lat: 46.877S

Lon: 37.859E

Elev: 24

StdP: 101.04

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	-1.0	-0.2	-6.9	2.1	1.7	-5.7	2.3	1.7	22.0	3.9	20.2	6.4	7.2	240	0.991

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.6	13.8	11.7	12.7	11.0	11.9	10.5	12.4	13.1	11.6	12.3	11.0	11.6	9.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	8.8	12.6	11.3	8.3	11.8	10.6	8.0	11.2	35.4	13.2	33.4	12.4	31.7	11.6	16.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.8	15.8	-2.8	17.3	0.9	1.7	-3.4	18.5	-3.9	19.6	-4.4	20.5	-5.0	21.8		
(5)			-3.8	14.4	0.8	1.2	-4.4	15.3	-4.9	16.0	-5.3	16.6	-5.9	17.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	8.0	8.3	7.7	6.8	5.8	4.8	4.5	4.2	4.1	5.0	5.9	7.1		
	DBStd	2.52	1.87	1.87	1.91	2.18	2.22	2.18	2.21	2.12	2.15	1.97	1.86	2.01		
	HDD10.0	1479	67	52	74	98	131	157	172	179	176	155	125	93		
	HDD18.3	4499	321	279	329	346	388	407	430	438	426	413	374	349		
	CDD10.0	21	5	6	4	3	1	0	0	0	0	1	1	2		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	7.9	7.6	7.2	7.5	7.5	7.9	7.9	8.6	8.3	8.5	7.9	7.7	7.8		
	PrecAvg	2221	194	179	187	191	199	184	174	167	175	161	167	185		
(15) Precipitation	PrecMax	2949	350	343	402	363	428	284	310	252	359	291	293	328		
	PrecMin	1429	85	73	16	45	60	91	74	87	79	75	68	11		
	PrecStd	373	60	62	71	63	69	47	46	47	52	49	48	61		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	15.4	14.3	14.5	12.9	10.9	11.0	9.9	10.7	11.9	12.8	14.3		
		MCWB	12.3	11.6	12.2	12.0	10.8	9.2	9.1	8.9	9.1	9.9	11.0	11.6		
	2%	DB	13.2	13.3	12.7	12.3	11.1	9.8	9.3	8.9	9.1	10.1	10.8	12.2		
		MCWB	11.6	11.1	11.2	11.0	10.0	8.8	8.4	8.1	7.9	8.7	9.2	10.2		
	5%	DB	12.1	12.2	11.7	11.2	10.1	9.0	8.6	8.2	8.1	9.0	9.7	11.2		
		MCWB	10.7	10.7	10.6	10.3	9.3	8.4	7.8	7.5	7.3	8.0	8.2	9.6		
	10%	DB	11.1	11.4	10.9	10.2	9.3	8.3	7.9	7.6	7.4	8.1	8.9	10.3		
		MCWB	9.8	10.2	9.8	9.4	8.5	7.6	7.2	6.9	6.6	7.2	7.7	9.0		
	0.4%	WB	13.7	13.2	12.9	12.6	11.8	10.1	9.7	9.4	9.8	10.2	11.8	12.6		
		MCDB	14.1	14.0	13.7	13.3	12.3	10.5	10.2	9.6	10.4	11.0	12.8	13.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	12.2	12.0	11.7	11.4	10.3	9.1	8.7	8.3	8.2	9.0	9.8	11.1		
		MCDB	12.7	12.5	12.3	12.0	10.8	9.5	9.1	8.6	8.6	9.7	10.4	11.7		
	5%	WB	11.1	11.2	10.9	10.5	9.5	8.5	8.1	7.7	7.5	8.2	8.7	10.2		
		MCDB	11.8	11.7	11.4	11.0	10.0	8.9	8.5	8.0	7.9	8.8	9.3	10.8		
	10%	WB	10.1	10.4	10.1	9.5	8.7	7.7	7.3	7.1	6.8	7.4	8.0	9.4		
		MCDB	10.8	11.1	10.7	10.2	9.2	8.2	7.8	7.6	7.3	8.0	8.7	10.1		
	5% DB	MDBR	4.5	4.6	4.3	4.1	4.0	3.8	3.8	3.9	4.0	4.2	4.4	4.4		
		MCDBR	6.4	6.1	5.5	5.4	5.5	4.8	4.6	4.5	4.8	5.3	5.5	5.7		
		MCWBR	5.7	5.5	5.5	5.0	5.7	5.0	4.8	4.9	4.9	4.9	4.9	4.9		
		MCWBR	6.1	5.6	5.4	5.2	5.3	4.6	4.6	4.4	4.7	4.9	5.1	5.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.9	5.6	5.7	5.1	5.8	5.2	5.0	5.0	5.1	5.0	5.1	5.0		
	taub		0.353	0.345	0.342	0.328	0.332	0.336	0.345	0.353	0.364	0.357	0.353	0.352		
		taud	2.489	2.525	2.523	2.539	2.407	2.309	2.307	2.355	2.347	2.420	2.458	2.467		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		964	941	890	820	708	639	672	767	846	917	958	972		
		Edh at Noon	110	100	90	74	67	65	73	86	105	110	113	114		
	RadAvg		5.24	4.44	3.17	1.92	1.21	0.84	0.99	1.67	2.68	3.81	4.83	5.49		
(44) All-Sky Solar Radiation	RadStd		0.41	0.26	0.16	0.13	0.09	0.05	0.05	0.11	0.15	0.29	0.34	0.47		

Historical Trends

DBAvg	Heating		Cooling			Degree-Days			
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	-0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page